Report Exercise 1 - Geomechanics and Structural Geology

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Part 1: Mohr-Coulomb Diagram

Task 1: Constructing the Mohr-Coulomb failure criteria for the following series of experiments on the Red Felser sandstone.

```
R^2 = 0.9644
0.9077552576785977
52.01054509579416
Cohesion (C) = 9.31 MPa

Mohr-Coulomb Parameters:
Cohesion (C) = 9.31 MPa
Friction Angle (ϕ) = 52.01°
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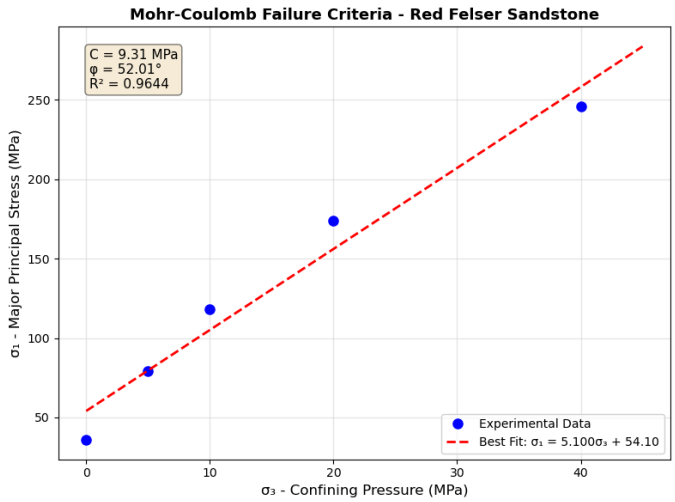


Based on the slope we claculated the friction angle based on this formula :

$$\sqrt{a} = \tan\left(45^\circ + \frac{\varphi}{2}\right)$$

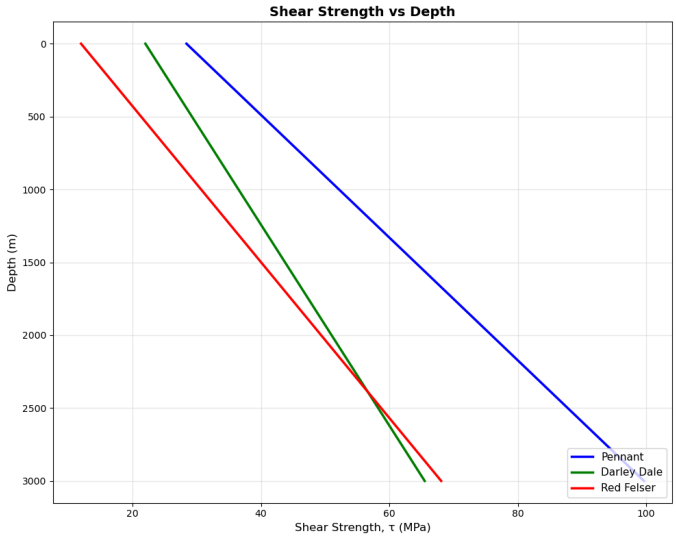
and based on the intercept we can derive the cohesion based on this formula:

$$b = 2C\tan\left(45^\circ + \frac{\varphi}{2}\right)$$

Based on the calculation, the cohesion (C) = 11.98 MPa and friction angle (φ) = 42.23°. The linear regression shows good fit (R^2 = 0.964) confirming the validity of Mohr-Coulomb criterion for this sandstone. We see that the samples become stronger as the confinement pressure increases as the sigma 1 (value at failure) increases.

Task 2: Constructing pressure vs shear strength plot with depth for the different rock types in thepresentation (e.g. Red Felser, Pennant, Darley Dale Sst. etc.).

Rock Type	C (MPa)	ϕ (°)	ρ (kg/m³)	porosity (%)
Pennant	28.40	43.53	2549	4.57
Darley Dale	22.00	32.62	2310	13.50
Red Felser	11.98	42.23	2100	21.40



Interpretation: lower porosity generally correlates with higher rock strength, as demonstrated by Pennant sandstone (4.57% porosity) exhibiting the highest cohesion (28.40 MPa) and steepest strength gradient with depth compared to Darley Dale (13.5%) and Red Felser (21.4%). Porosity also controls density and frictional behavior, which together determine how rapidly shear strength increases with depth. As a result, Pennant's low porosity and high density make it far more responsive to confining stress, whereas the highly porous Red Felser remains mechanically weaker even at great depth.

Task 3(a): Evaluating what happens when the Poisson's ratio is variable.

$$\sigma_3(z) = \frac{\nu}{1-\nu}\rho g z$$

Stress Ratio:

$$\sigma 3/\sigma 1 = \nu/(1-\nu)$$

If ν increases:

- The ratio σ3 / σ1 increases.
- The horizontal stress σ3 becomes closer to the vertical stress σ1
- The differential stress (σ1 − σ3) decreases.

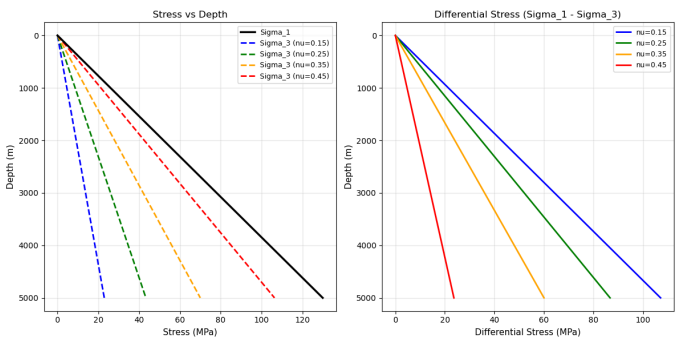
If ν decreases:

- The ratio σ3 / σ1 decreases.
- The horizontal stress becomes smaller relative to the vertical stress.
- The differential stress (σ1 − σ3) increases.

This means that for very small ν (brittle, highly compressible rock which does not expand a lot on the horizontal direction) σ3 is much smaller than σ1, the differential stress (σ1 − σ3) is big so the rock reaches quicker the fail point because it cannot expand on the sides.

For a big ν (rock which deforms a lot more on the horizontal direction when a vertical stress is applied), the rock will resist much better to a vertical load because it can expand so the failure point comes much later.

A simulation under different numbers of Poission's ratio is performed as follow:

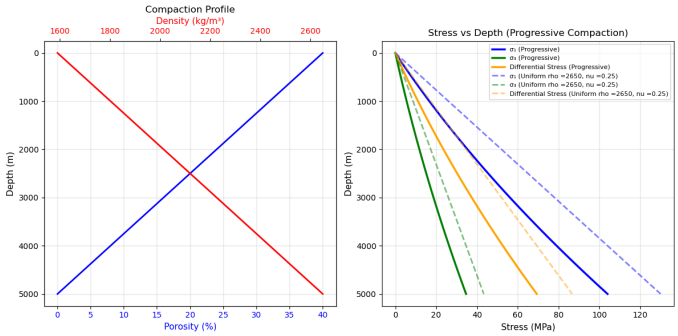


Interpretation: higher Poisson's ratio leads to higher horizontal stress (sigma 3) and lower differential stress, making rock less prone to failure. This demonstrates that brittle rocks (low Poisson's ratio) experience larger differential stress and reach failure more easily than ductile rocks (high Poisson's ratio) under the same vertical load.

Task (3b): Evaluating what happens when the porosity is decreasing from 40% at the Earth Surface to 0% at 5 km depth under the assumption the whole system contains of sandstones (density of non-porous sandstone is 2.65 g/cm³).

Progressive compaction creates depth-dependent density, requiring the average density method to properly account for the varying overburden weight. This differs from assuming 0% porosity everywhere, which overestimates the actual stress.

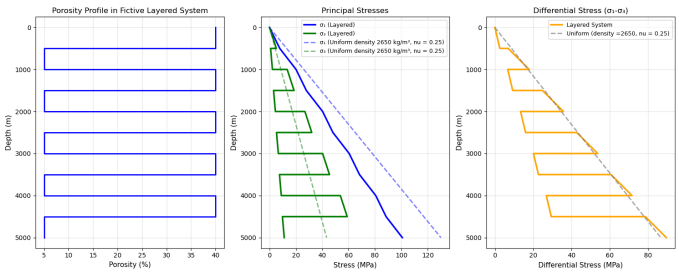
A simulation of progressive compaction is performed as follow:



Interpretation: the progressive compaction model (porosity 40%→0%) shows that vertical stress (σ_1) reaches 103.96 MPa at 5 km depth. Horizontal stress (σ_3) = $\sigma_1/3$ following normal faulting regime with ν=0.25. Left panel shows the compaction profile: porosity decreases linearly while density increases from 1590 kg/m³ (surface) to 2650 kg/m³ (5 km depth). Compared to uniform rock baseline (dashed lines: ρ=2650 kg/m³, ν=0.25), the *progressive compaction scenario produces lower stresses* throughout most of the depth due to lower average density.

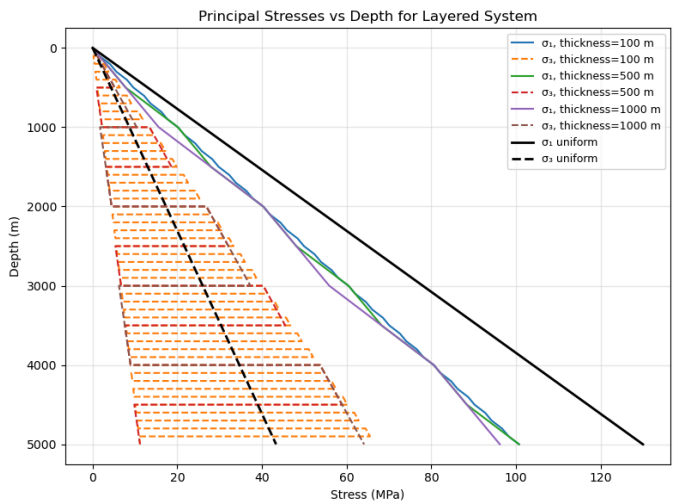
Task (3c): Evaluating what happens when a fictive layered system with significant variations in porosity (and thus density) and Poisson's ratio incorporated (high porosity layers, e.g. 40% with Poisson's ratio ~0.4 vs low porosity layers, e.g <5% with low Poisson's ratio ~0.1). Also variating layer thickness to see how the sigma 1, sigma 3 vary.

The simulation below creates alternating layers and calculates cumulative stress accounting for overburden from all previous layers.



Interpretation: the layered system shows a "staircase" pattern with stress jumps at boundaries due to varying Poisson's ratios - high porosity layers (ν=0.4) produce lower differential stress in average, while low porosity layers (ν=0.1) generate higher differential stress in average.

If we would play with diffferent layer thickness we would observe this :



While varying the layer thickness we observe that thin layers increase local stress contrasts and produce strong oscillations in σ_1 and σ_3, whereas thicker layers smooth these variations and make the system behave more like a uniform medium. As a result, differential stress becomes increasingly controlled by bulk density and Poisson's ratio as layer thickness increases.

Task 4: Now we know the difference between sigma 1 and 3 for the different scenarios mentioned above and see how far or how much above we get to the Mohr-Coulomb criteria for the different lithologies. Are we expecting any failure and fracturing in any of the scenarios or not? Use an extensional stress regime for the analyses and a dominant fault/fracture orientation of 70° from sigma 3.

For this we will analyse the three different cases and for each, where the differential stress is maximum because this is where the probabilty of failure is the highest as well.

<u>For case *a* - uniform density :</u>

To expect if there is any failure and fracturing, we focus on the highest depth (5 km) and where the poisson ration is the lowest (ν= 0.15) because if there is no failure in this case, it will normaly not fail for the other cases.

The fault as an orientation of 70° to σ_3 and we know that there is 90° between σ_3 and σ1 so there is 20° form σ1 to the fault.

- Density: $\rho = 2650\ \mathrm{kg/m^3}$
- Vertical stress: $\sigma_1 = \rho g z \approx 130\ \mathrm{MPa}$
- $\sigma_3 = \frac{\nu}{1-\nu}\sigma_1$
- Minimum horizontal stress:

$$\sigma_3 = \frac{0.15}{0.85} \times 130 \approx 22.9\ \mathrm{MPa}$$

- Formula of the normal stress on the fault plane:

$$\sigma_n = \frac{\sigma_1 + \sigma_3}{2} + \frac{\sigma_1 - \sigma_3}{2}\cos(2\alpha)$$

- Shear stress on the fault plane:

$$\tau = \frac{\sigma_1 - \sigma_3}{2}\sin(2\alpha)$$

- Mohr-Coulomb failure criterion:

$$\tau_{\text{fail}} = c + \sigma_n \tan\varphi$$

- Normal stress:

$$\sigma_n = 76.5 + 53.5 \cdot 0.766 \approx 117.5\ \mathrm{MPa}$$

- Shear stress:

$$\tau = 53.5 \cdot 0.643 \approx 34.4\ \mathrm{MPa}$$

Mohr-Coulomb Criterion for Each Rock:

we had,

Rock Type	c **(MPa)**	φ **(°)**	$\tan\varphi$
Pennant	28.40	43.53	0.952
Darley Dale	22.00	32.62	0.639
Red Felser	11.98	42.23	0.904

<u>Pennant Sandstone</u>

$$\tau_{\text{fail}} = 28.40 + 117.5 \times 0.952 \approx 28.40 + 111.9 = 140.3\ \mathrm{MPa}$$

$\tau = 34.4\ \mathrm{MPa} \ll 140.3\ \mathrm{MPa}$
➡️ Far from failure

<u>Darley Dale Sandstone</u>

$$\tau_{\text{fail}} = 22 + 117.5 \times 0.639 \approx 22 + 75.1 = 97.1\ \mathrm{MPa}$$

Comparison:
$\tau = 34.4\ \text{MPa} < 97.1\ \text{MPa}$
➡️ Weakest among the three, but still no failure

Red Felser Sandstone

$$\tau_{\text{fail}} = 11.98 + 117.5 \times 0.904 \approx 11.98 + 106.3 = 118.3\ \text{MPa}$$

Comparison:
$\tau = 34.4\ \text{MPa} < 118.3\ \text{MPa}$
➡️ No failure

For case *b* - Progressive compaction:

Again, we will calculate for a depth of 5 km where σ1 is the highest, if there is no failure in this case, it will normaly not fail for the other cases.

For progressive compaction (porosity 40% → 0%) at 5 km: $\sigma_1(5\ \text{km}) \approx 103.96\ \text{MPa} \approx 104\ \text{MPa}$

We keep $\sigma_1 \approx 104\ \text{MPa}$ and $\nu = 0.25$

- Normal faulting regime:

$$\sigma_3 = \frac{\nu}{1-\nu}\sigma_1 = \frac{0.25}{0.75}\sigma_1 = \frac{1}{3}\sigma_1$$

$$\sigma_3 \approx \frac{104}{3} \approx 34.65\ \text{MPa}$$

- Normal stress:

$$\sigma_n = 69.35 + 34.65 \cdot 0.766$$

$$\sigma_n \approx 95.9\ \text{MPa}$$

- Shear stress:

$$\tau = 34.65 \cdot 0.643$$

(see formula above)

$$\tau \approx 22.3\ \text{MPa}$$

Mohr–Coulomb Criterion for the Same Rocks:

Pennant Sandstone

$$\tau_{\text{fail}} = 28.40 + 95.9 \cdot 0.952 \approx 28.40 + 91.3 = 119.7\ \text{MPa}$$

Comparison:
$\tau = 22.3\ \text{MPa} < 119.7\ \text{MPa}$
➡️ Far from failure

Darley Dale Sandstone

$$\tau_{\text{fail}} = 22 + 95.9 \cdot 0.639 \approx 22 + 61.3 = 83.3\ \text{MPa}$$

Comparison:
$\tau = 22.3\ \text{MPa} < 83.3\ \text{MPa}$
➡️ No failure

Red Felser Sandstone

$$\tau_{\text{fail}} = 11.98 + 95.9 \cdot 0.904 \approx 11.98 + 86.7 = 98.7\ \text{MPa}$$

Comparison:

$\tau = 22.3 \text{ MPa} < 98.7 \text{ MPa}$
➡️ No failure

<u>For case c - layered system with variations in porosity:</u>

We will again consider the layer at 5km depth and the most critical ones are with low porosity (<5%) because higher density so higher σ1 and low poisson's ratio (0.1)

From the stress plot: $\sigma_1^{\text{layered}}(5 \text{ km}) \approx 100 \text{ MPa}$

$$\sigma_3 = \frac{\nu_{\text{layer}}}{1 - \nu_{\text{layer}}} \sigma_1$$

With $\nu_{\text{layer}} = 0.1$ and $\sigma_1 = 100 \text{ MPa}$:

$$\sigma_3 = \frac{0.10}{1 - 0.10} \cdot 100 = \frac{0.10}{0.90} \cdot 100$$

$$\sigma_3 \approx 11.1 \text{ MPa}$$

$$\sigma_n = 55.6 + 44.4 \times 0.766$$

$$\sigma_n \approx 89.7 \text{ MPa}$$

Shear Stress on the Plane:

$$\tau = \frac{\sigma_1 - \sigma_3}{2} \sin(40^\circ)$$

$$\tau \approx 28.6 \text{ MPa}$$

Mohr–Coulomb Criterion for the Same Rocks:

<u>Pennant Sandstone</u>

$$\tau_{\text{fail}} = 28.40 + 89.6 \cdot 0.952$$

$$\tau_{\text{fail}} \approx 113.7 \text{ MPa}$$

Comparison:

$\tau = 28.6 \text{ MPa} < 113.7 \text{ MPa}$

➡️ Far from failure

<u>Darley Dale Sandstone</u>

$$\tau_{\text{fail}} = 22 + 89.6 \cdot 0.639$$

$$\tau_{\text{fail}} \approx 79.2 \text{ MPa}$$

Comparison:

$\tau = 28.6 \text{ MPa} < 79.2 \text{ MPa}$

➡️ No failure

<u>Red Felser Sandstone</u>

$$\tau_{\text{fail}} = 11.98 + 89.6 \cdot 0.904$$

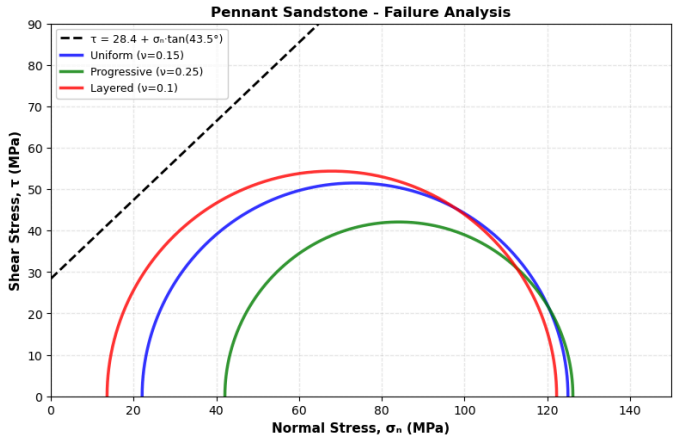
$$\tau_{\text{fail}} \approx 93.0 \text{ MPa}$$

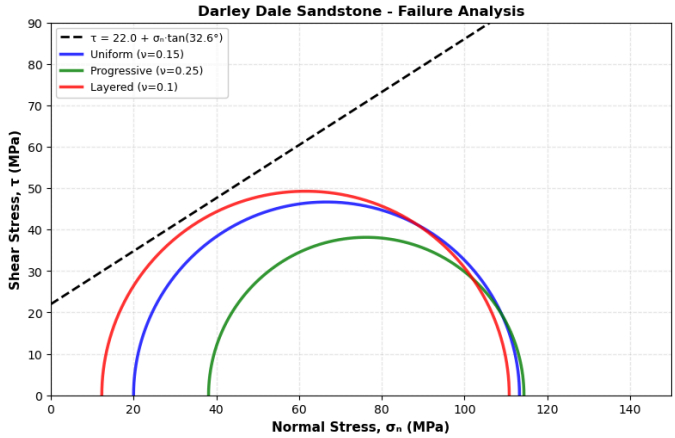
Comparison:

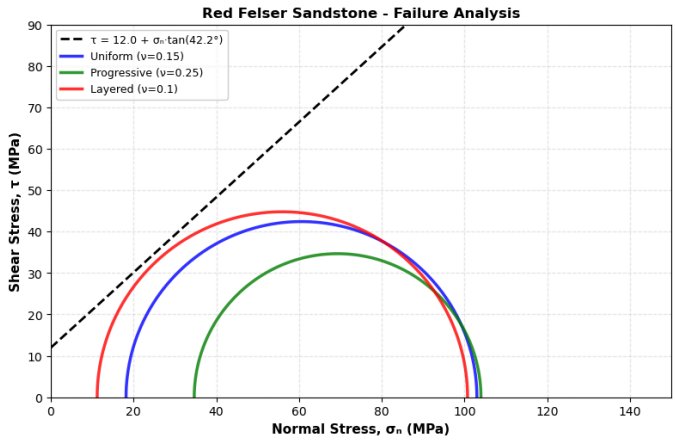
$\tau = 28.6 \text{ MPa} < 93.0 \text{ MPa}$

➡️ Closer to failure but still no failure

An illustration of the case is provided in the following plot:







Interpretation: Each diagram shows one rock type evaluated under three geological scenarios (uniform, progressive compaction, layered system), with all Mohr circles remaining safely below their respective failure envelopes indicating no failure is expected. The uniform scenario produces the largest differential stress due to low Poisson's ratio (ν=0.15), while the layered system shows the smallest circles due to lower average density. Pennant sandstone demonstrates the highest failure resistance (C=28.4 MPa, φ=43.5°), followed Darley Dale (C=22.0 MPa, φ=32.6°) and Red Felser (C=11.98 MPa, φ=42.2°).

Task 5: Minimum fluid pressure introduction to lead failure in some scenarios

When fluid pressure (Pf) is applied, effective stresses are reduced:

$$\sigma_1' = \sigma_1 - P_f$$

$$\sigma_3' = \sigma_3 - P_f$$

$$P_f = \frac{\sigma_1 - \sigma_3 \cdot N_\varphi - 2C\sqrt{N_\varphi}}{1 - N_\varphi}$$

Where $N_\varphi = \frac{1+\sin\varphi}{1-\sin\varphi} = \tan^2\left(45^\circ + \frac{\varphi}{2}\right)$

Illustrations of fluid pressure for failure analysis are showed here:

```
==============================================================================
Pennant Sandstone - Fluid Pressure Analysis
==============================================================================
Uniform (ν=0.15)          σ₁(total)=  125.03  σ₃(total)=   22.06  Pf=   28.69
Progressive (ν=0.25)      σ₁(total)=  126.22  σ₃(total)=   42.07  Pf=   52.95
Layered (ν=0.1)           σ₁(total)=  122.29  σ₃(total)=   13.59  Pf=   18.92
==============================================================================
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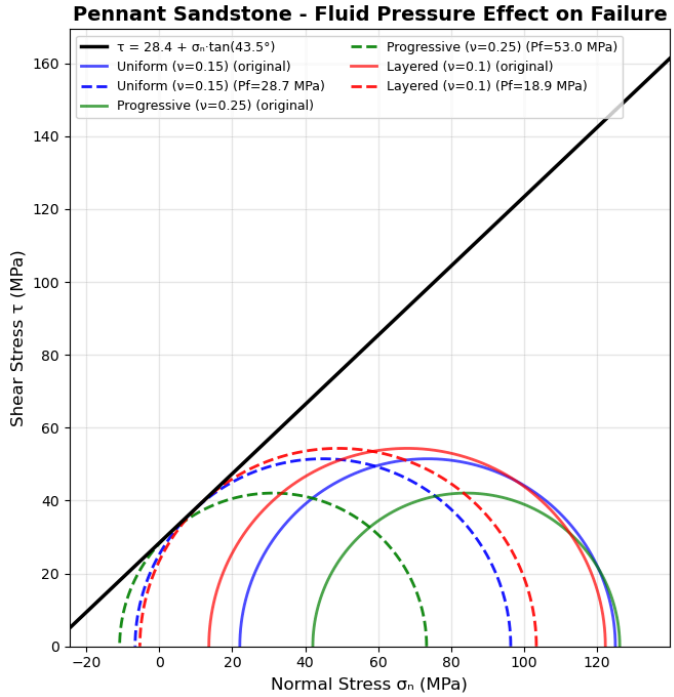
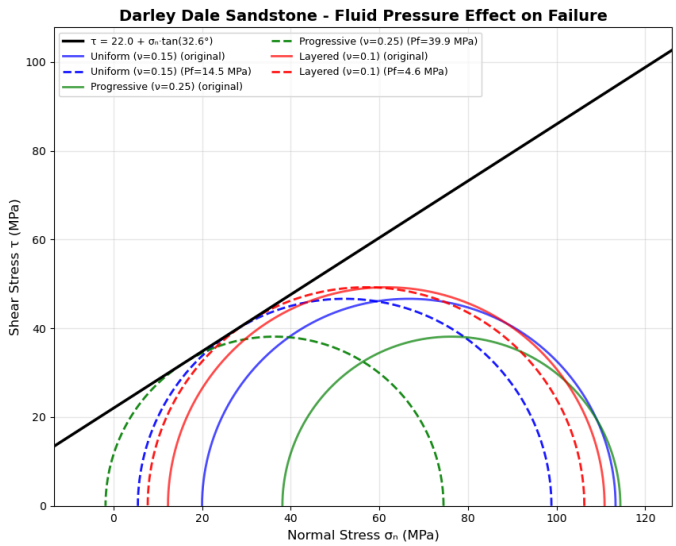


```
==============================================================================
Darley Dale Sandstone - Fluid Pressure Analysis
==============================================================================
Uniform (ν=0.15)          σ₁(total)=  113.31  σ₃(total)=   20.00  Pf=   14.48
Progressive (ν=0.25)      σ₁(total)=  114.38  σ₃(total)=   38.13  Pf=   39.90
Layered (ν=0.1)           σ₁(total)=  110.82  σ₃(total)=   12.31  Pf=    4.57
==============================================================================
```



```
==============================================================================
Red Felser Sandstone - Fluid Pressure Analysis
==============================================================================
Uniform (ν=0.15)          σ₁(total)=  103.00  σ₃(total)=   18.18  Pf=   10.68
Progressive (ν=0.25)      σ₁(total)=  103.99  σ₃(total)=   34.66  Pf=   30.95
Layered (ν=0.1)           σ₁(total)=  100.75  σ₃(total)=   11.19  Pf=    2.55
==============================================================================
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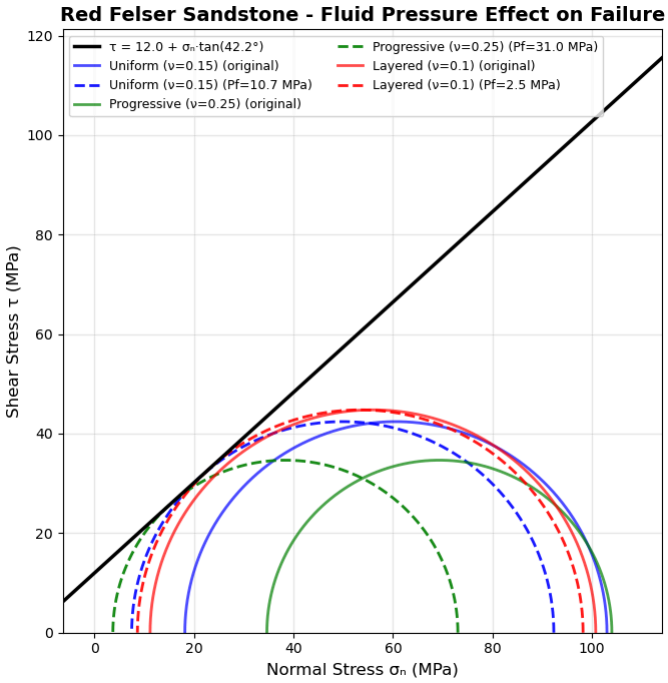

Interpretation: Each diagram visualizes how increased fluid pressure shifts the Mohr circle leftward (reducing effective stresses) until it becomes tangent to the failure envelope, with solid circles showing original stress states and dashed circles showing critical failure conditions. Red Felser requires the lowest fluid pressure to induce failure (7-19 MPa) due to its low cohesion, while Pennant demonstrates highest resistance (40-60 MPa), with the uniform scenario consistently requiring higher Pf values across all rock types. These critical fluid pressures define the upper limit of the mud weight window for safe drilling operations - exceeding these values would cause formation failure.

Part 2: Linear Elastic Fracture Mechanics

Task 1: Creating maximum stress distribution and stress intensity plots away from a hole for a typical average stress level at 3 km depth (using information from the 1st set of exercises) under different fracture geometries.

```
At 3 km: sigma_v = 77.99 MPa, sigma_h = 26.00 MPa
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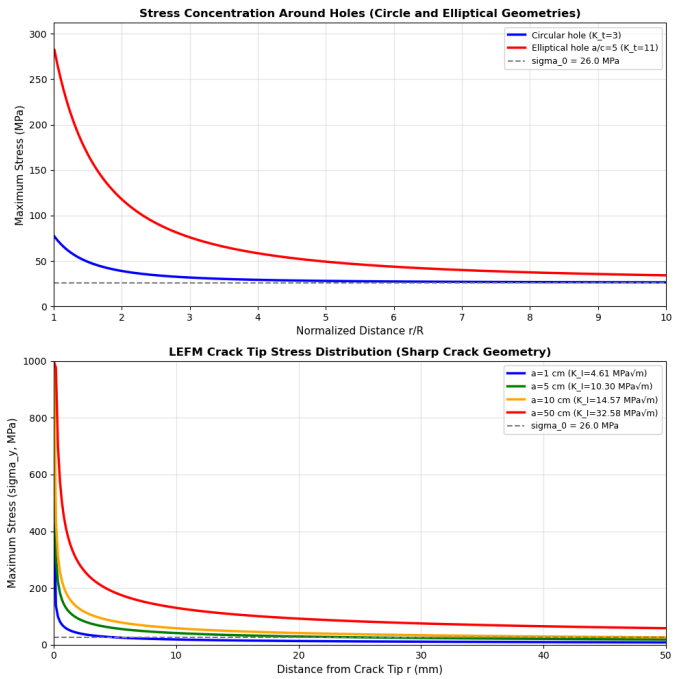


Interpretation: The plots reveal stress amplification differences between fracture geometries at 3 km depth (σ_0 = 26 MPa).

The upper plot depicts the stress concentration around holes (where r is actual distance and R is hole radius, and r/R represents normalized distance from hole center). At $r/R = 1$ (at hole edge), the maximum stress occurs here. At $r/R = 2 - 10$ (moving away from hole), stress decays back to σ_0 (far-field stress). Afterall, circular hole indicates moderate stress concentration (K_t = 3, σ_max ≈ 78 MPa at edge), while elliptical holes (a/c=5) shows bigger stress concentration (K_t = 11, σ_max ≈ 286 MPa at tip).

The lower plot depicts sharp crack tip stress field (distance from crack tip):. The r is absolute distance from crack tip in millimeters, that it shows LEFM's 1/√r singularity (stress is highest near crack tip). For this type of crack, a longer crack produces more intense stress fields, for instance the 50 cm crack has ~1000 MPa at r = 0.1 mm from tip, while 1 cm crack has ~300 MPa at r = 0.1 mm from the tip.

Task 2: What type of fracture geometry would you need to have stress levels at the tip of the fracture in excess of the Mohr-Coulomb failure streng.

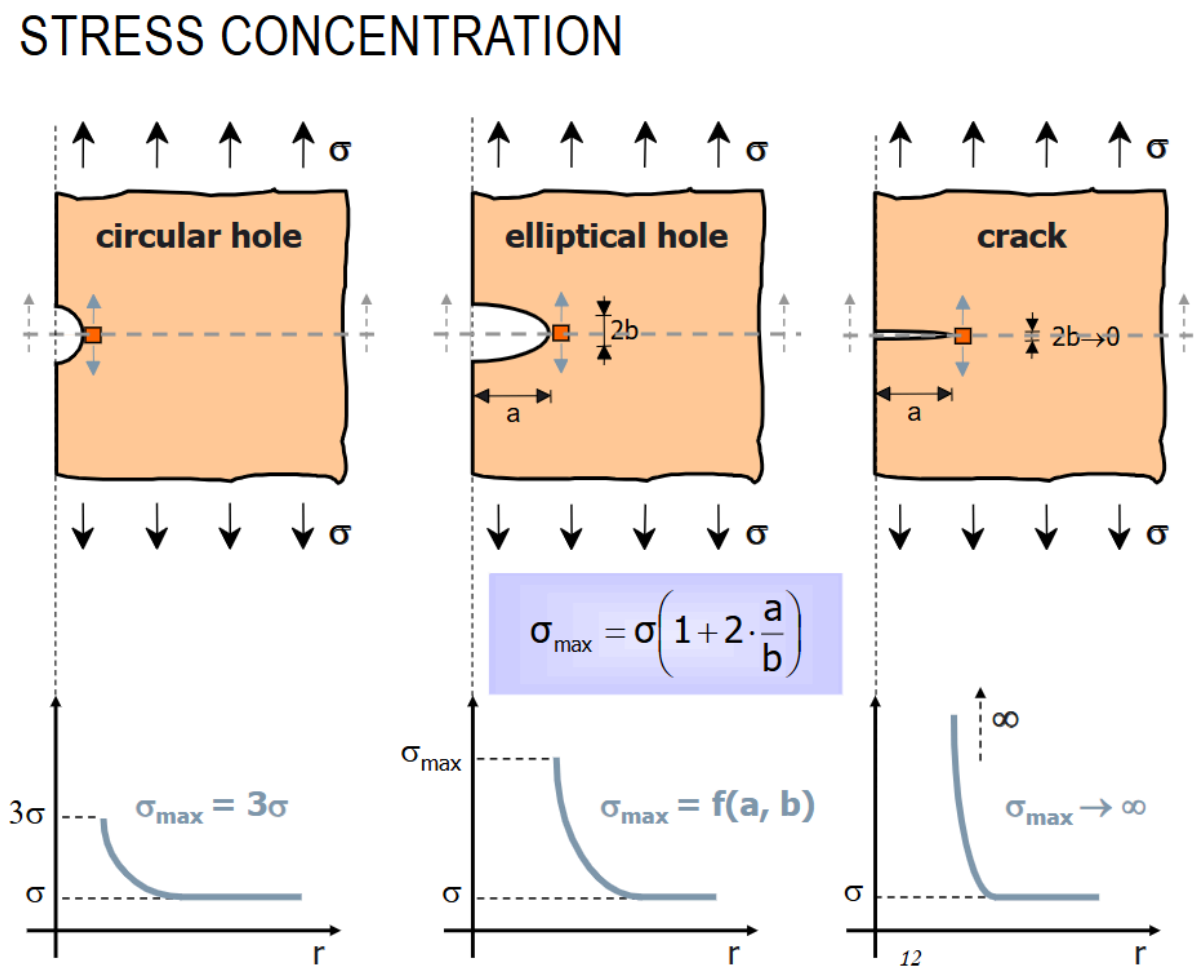
Based on Task 1 of Part 2, to obtain stress levels at the tip of the fracture that exceed the Mohr-Coulomb failure strength :

$$\tau = c + \sigma_n \tan\phi$$

the fracture must have a very sharp geometry. Circular holes (at the very left side of the image bellow) only produce moderate stress concentration (K_t = 3) which in general remains bellow the rock failure strength. In contrast, elliptical fractures (figure in the middle of the image bellow) with a large a/c ratio increases the stres concentration at the tip and in the case where the minor axis c approaches 0, the geometry becomes sharp (on the very right side of the image). Stresses near the a sharp crack tip increases as

$$\sigma \propto \frac{1}{\sqrt{r}}$$

leading to very large local stresses. Therefore sharp ckracks or very ellongated elliptical fractures are reuqired to generate stresses that exceed the Mohr-Coulomb failure strenght and allow fracture propagation.


STRESS CONCENTRATION
σ
circular hole
elliptical hole
crack
2b
2b→0
a
σ
σ
σ

- a = radius of the long axis of the fracture

For plane strain, the relationship between energy release rate G_I and stress intensity factor K_I is:

$$G_I = \frac{K_I^2}{E'} = K_I^2 \left(\frac{1-\nu^2}{E} \right)$$

where:

$$E' = \frac{E}{1-\nu^2}$$

Derivating critical stress, fracture propagation occurs when:

$$G_I = 2\gamma$$

Therefore:

$$\frac{K_I^2}{E'} = 2\gamma$$

Using $K_I = \sigma\sqrt{\pi a}$ and substituting:

$$\frac{\sigma^2 \pi a}{E'} = 2\gamma$$

This gives the critical stress with the converted Young's modulus:

$$\sigma_{\text{crit}} = \sqrt{\frac{2\gamma E'}{\pi a}}$$

We assume fracture energy $\gamma = 100\ \mathrm{J/m^2}$

Application to Red Felser Sandstone:

Mode I corresponds to opening of the fracture perpendicular to the fracture surface. For a vertical fracture, it's the horizontal stress σ_h that causes the opening. We use the normal stress calculated for Red Felser sandstone:

$$\sigma_h = 26\ \mathrm{MPa}$$

We conclude that:

- If $\sigma_{\text{crit}} > 26$ MPa: fracture does not propagate
- If $\sigma_{\text{crit}} \leq 26$ MPa: fracture can propagate

For different fracture radii, we compute the graph below:

```
E' = 16.0 GPa

Results :
a = 0.1 cm  ->  sigma_crit = 32.96 MPa
a = 0.5 cm  ->  sigma_crit = 14.74 MPa
a = 1.0 cm  ->  sigma_crit = 10.42 MPa
a = 5.0 cm  ->  sigma_crit = 4.66 MPa
a = 10.0 cm  ->  sigma_crit = 3.30 MPa
a = 50.0 cm  ->  sigma_crit = 1.47 MPa
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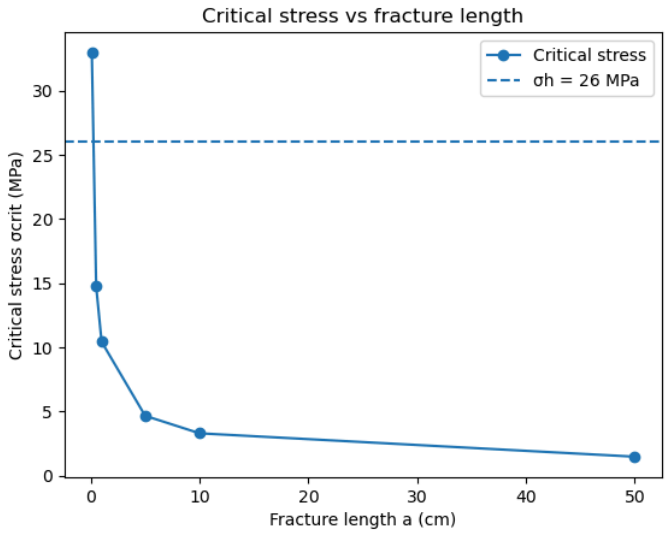


Interpretation: : We notice that when the fracture elongates, the critical stress (stress required for the fracture to propagate) deacreses. Also, if σ h > σ critical, the fracutures can propagate and we notice that this is the case for most fractures lenght. So in the conditions of stress of the Red Felser, most type of fracutres can propagate execpt for a very tiny fractures(0,1 cm).

Task 4 (Bonus Part): Making a stress concentration plot aroundthe fracture tip.

A stress concentration plot around a fracture tip was generated as a reference for future finite element model (FEM) validation. The analysis is based on Mode I linear elastic fracture mechanics (LEFM), with far-field loading from Part 2 Task 1:

$$\sigma_0 = \sigma_h = 26 \text{ MPa}$$

For pure opening mode ($K_{II} = 0$), the near-tip stresses are:

$$\sigma_r = \frac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left(1 + \sin^2\frac{\theta}{2}\right)$$

$$\sigma_\theta = \frac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\cos^2\frac{\theta}{2}$$

$$\tau_{r\theta} = \frac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\sin\theta$$

with

$$K_I = \sigma_0\sqrt{\pi a}$$

The opening stress used for interpretation is the Cartesian component σ_y after transforming from polar to Cartesian coordinates.

```
Crack parameters:
  Half-length: 5.0 cm
  Applied stress: 26.0 MPa
  Stress intensity K_I: 10.30 MPa√m
```

Step 1: Stress Field Calculation

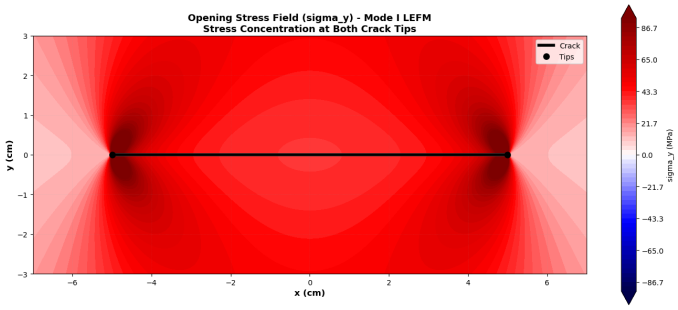
The stress field was computed on a 2D grid by superposing the contribution of both crack tips. The opening stress component (σ_y) was retained because it controls Mode I crack opening.

```
Full crack stress field calculated:
  Crack: -5.0 cm to 5.0 cm (length = 10.0 cm)
  Max opening stress (sigma_y): 271.0 MPa
  (Occurs at theta=0, directly ahead of crack tips)
```

Step 2: Stress Concentration Visualization

A contour map of σ_y was produced to show stress concentration at the crack tips and decay away from the fracture. The spatial trend follows the LEFM scaling:

$$\sigma \propto \frac{1}{\sqrt{r}}$$



Interpretation: The stress field is symmetric about the crack and shows maximum opening stress directly ahead of both tips. This is consistent with Mode I behavior and identifies the tip regions as the most likely zones for fracture propagation.